ABB NEXUS : Maintenance Intelligence

Multimodal Maintenance Intelligence Platform — Project Understanding & Dataset Evaluation

Event: ABB Hackathon 2026 | **Architecture:** InduSense (Predict-Retrieve-Align-Validate) | **Focus:** Grounded AI for Industrial Troubleshooting | **Date:** October 2026

1. Executive Summary & Project Understanding

ABB Nexus is an enterprise-grade **Multimodal Maintenance Intelligence Platform** designed to eliminate costly unplanned machine downtime by bridging two historically isolated domains:

1. **Industrial Sensor Telemetry:** High-frequency time-series signals from sensors (temperature, RPM, torque, tool wear, vibration) used by Deep Learning / ML models to classify fault modes and predict Remaining Useful Life (RUL).
2. **Technical Document Intelligence:** Thousands of pages of unstructured technical manuals, electrical schematics, troubleshooting matrices, and historical work orders.

> **The Problem with Existing Solutions:** Standard prediction models only output abstract labels like "*likely bearing failure*" without indicating what maintenance steps must be performed. Conversely, generic PDF chatbots or RAG systems answer queries based on semantic keyword matching, frequently returning procedures that sound similar but are technically inapplicable to the machine's actual operating condition.

The ABB Nexus Core Novelty: Integrating the **InduSense** framework — a 6-stage grounded architecture that executes: **Predict** → **Retrieve** → **Align** → **Validate** → **Adapt** → **Explain**. The system computes a *Condition-to-Evidence Alignment (CEA) score*, fusing telemetry confidence with document evidence confidence to verify that the retrieved procedure strictly applies to the predicted physical fault.

2. System Pipeline Architecture

1. PREDICT (ML / DL)

Ingests telemetry CSV streams. XGBoost & deep models predict condition labels (TWF, HDF, PWF, OSF) and calculate prediction confidence P_C.

2. RETRIEVE (RAG)

Ingests manuals using OCR, layout parsing, and table extraction. Retrieves candidate passages and outputs retrieval confidence R_C.

3. ALIGN (CEA Score)

Calculates Condition-to-Evidence Alignment matching component → fault → symptom → action rather than blind semantic text similarity.

4. VALIDATE (Fusion)

Fuses P_C, R_C, and CEA into a unified reliability metric. Automatically flags prediction-evidence conflicts.

5. ADAPT (Compute)

Routes inference between lightweight edge execution (high confidence) and heavy cloud/expert review (conflicts or low confidence).

6. EXPLAIN (Evidence)

Presents grounded answers with exact document IDs, page numbers, bounding box highlights, and confidence metrics to technicians.

3. Dataset Master Inventory & Recommendation Matrix

An exhaustive review of all 12 folders inside your local `Dataset` directory was conducted. The table below categorizes each dataset by its operational tier, target system module, and suitability for the ABB Hackathon platform.

Folder Name in Dataset/	Identified Dataset	Tier	Target System Role	Actionable Status
ai4i+2020+predictive+maintenance+dataset	UCI AI4I 2020 Predictive Maintenance	TIER 1: CORE	Telemetry Fault Mode Classifier (XGBoost / LightGBM)	**Active / Primary** (Preprocessed in repo)
CMAPSSData (1) & **6.+Turbofan...**	NASA C-MAPSS Turbofan Engine Degradation	TIER 1: CORE	Remaining Useful Life (RUL) & Degradation Trajectory	**Active / Primary** (Standard benchmark)
archive (59)	FUNSD (Form Understanding in Noisy Scanned Docs)	TIER 1: CORE	Document AI: Noisy OCR & Scanned Form Parsing	**Active / Primary** (199 annotated forms)
PubLayNet-master	IBM / ICDAR PubLayNet	TIER 1: CORE	Document AI: Layout Parsing (Text, Titles, Figures, Tables)	**Active / Primary** (Preserves manual layout)
TableBank-master	Microsoft TableBank	TIER 1: CORE	Document AI: Troubleshooting & Lookup Matrix Extraction	**Active / Primary** (Matrix & table parser)
10.+FEMTO+Bearing	PRONOSTIA / FEMTO Bearing Run-to-Failure	TIER 2: HIGH	Vibration & Thermal Bearing Degradation Diagnostics	**High-Value Showcase** (Rotating equipment)
3.+Milling	NASA Milling Dataset	TIER 2: HIGH	CNC Tool Wear Progression (Acoustic & Current signals)	**High-Value Showcase** (Complements TWF in AI4I)
archive (64)	SROIE 2019 (+ LayoutLM Base uncased)	TIER 2: HIGH	Work-Order Entity Extraction & LayoutLM Weights	**Useful Asset** (Pretrained model weights)
11.+Randomized+Battery+Usage...	NASA Randomized Battery Usage Data Set	TIER 3: OPT	Fleet Battery State-of-Health & Industrial UPS	**Optional Extension** (Specialized use case)
archive (60)	IIIT-5K Word Recognition V3.0	TIER 3: LOW	Isolated Word Recognition	**Low Priority** (Redundant with FUNSD/SROIE)

Folder Name in Dataset/	Identified Dataset	Tier	Target System Role	Actionable Status
1.+Algae+Raceway	NASA Algae Raceway Data Set	**TIER 4: SKIP**	Biochemical Algae Growth in Outdoor Ponds	**Do Not Use** (Out of industrial scope)

> **Key Insight on Repository Synergy:** Your codebase already has pre-configured dataset directories and model scripts inside `abb-maintenance-intelligence/ml/` that directly correspond to **AI4I**, **CMAPSS**, **FUNSD**, **PubLayNet**, and **TableBank**.

4. In-Depth Analysis of Individual Datasets

1. AI4I 2020 Predictive Maintenance Dataset

TIER 1: ESSENTIAL

Location: `Dataset/ai4i+2020+predictive+maintenance+dataset/ai4i2020.csv` (10,000 synthetic industrial records)

Parameters & Features: Air Temperature [K], Process Temperature [K], Rotational Speed [rpm], Torque [Nm], Tool Wear [min]. Identifies five specific machine failure modes:

- **TWF (Tool Wear Failure):** Tool wear exceeds critical limits (200 - 240 mins).
- **HDF (Heat Dissipation Failure):** Temp difference between air and process falls below 8.6 K and speed < 1380 rpm.
- **PWF (Power Failure):** Product of torque and rotational speed outside required operational envelope.
- **OSF (Overstrain Failure):** Product of tool wear and torque exceeds physical capacity.
- **RNF (Random Failure):** Process anomalies with 0.1% baseline occurrence probability.

Project Role: Powers your live Machine Health Dashboard, XGBoost classification, and provides the condition label that initiates document retrieval.

2. NASA C-MAPSS Turbofan Engine Degradation Simulation

TIER 1: ESSENTIAL

Location: `Dataset/CMAPSSData (1)/` & `Dataset/6.+Turbofan...` (FD001, FD002, FD003, FD004)

Parameters & Features: 3 operational settings (altitude, Mach number, throttle resolver angle) and 21 sensor channels (temperatures, pressures, fan speeds, bleed ratios) recorded from healthy initial state until system degradation and failure.

Project Role: The benchmark standard for Remaining Useful Life (RUL) regression in industrial PHM. Enables your platform to predict exact cycles/hours remaining before maintenance must occur.

3. FUNSD (Form Understanding in Noisy Scanned Documents)

TIER 1: ESSENTIAL

Location: `Dataset/archive (59)/` (199 scanned documents with 31,485 word-level annotations)

Parameters & Features: Real scanned document images with low-resolution scans, skew, stamps, and signatures. Full JSON annotations detailing word boxes, labels (header, question, answer, other), and entity relation links.

Project Role: Crucial for the OCR and document preprocessing pipeline, ensuring high extraction accuracy when technicians upload scanned physical inspection sheets and field service logs.

4. Microsoft TableBank & IBM PubLayNet TIER 1: ESSENTIAL

Location: `Dataset/TableBank-master/` and `Dataset/PubLayNet-master/`

Project Role: *PubLayNet* classifies complex PDF pages into text blocks, titles, figures, and tables. *TableBank* segments and parses table matrices. In industrial manuals, failure codes and torque parameters live inside lookup tables. These models prevent tabular data from turning into jumbled text chunks during RAG embedding.

5. FEMTO / PRONOSTIA Bearing Degradation Dataset

TIER 2: HIGHLY VALUABLE (DOMAIN EXTENSION)

Location: `Dataset/10.+FEMTO+Bearing/FEMTOBearingDataSet.zip`

Parameters & Features: High-frequency vibration signals (horizontal and vertical accelerometers at 25.6 kHz) and thermal telemetry collected on rotating ball bearings under varying radial loads and rotating speeds until complete degradation.

Project Role: Bearing failure is the single most common failure mode in ABB electric motors and drives. Incorporating FEMTO demonstrates vibration spectrum analysis (FFT, envelope analysis) alongside standard SCADA signals.

6. NASA Milling Dataset

TIER 2: HIGHLY VALUABLE (MACHINING EXTENSION)

Location: `Dataset/3.+Milling/mill.zip`

Parameters & Features: 167 manufacturing runs on a Matsuura machining center under diverse feed rates, cut depths, and materials. Measures acoustic emissions (table & spindle), vibration, and AC/DC drive currents to monitor progressive flank wear.

Project Role: Ideal validation dataset for physical tool wear telemetry, providing high-resolution physical waveforms that substantiate AI4I's tool wear metrics.

7. SROIE 2019 (Receipt / Work-Order Extraction + LayoutLM)

TIER 2: HIGHLY VALUABLE (MODEL WEIGHTS)

Location: `Dataset/archive (64)/`

Project Role: Contains pre-trained weights for `layoutlm-base-uncased`. LayoutLM fuses 2D spatial coordinates with text tokens, significantly improving information extraction from semi-structured maintenance work orders.

8. NASA Randomized Battery Usage & IIIT-5K

TIER 3: SECONDARY / OPTIONAL

Battery Usage (`Dataset/11.+Randomized...`): High quality electrochemical cycle data for lithium-ion cells. Useful if showcasing industrial Battery Energy Storage Systems (BESS) or automated guided vehicles (AGVs), but secondary for machinery maintenance.
IIIT-5K (`Dataset/archive (60)/`): Scene text word images. Redundant because FUNSD and SROIE cover full-page document OCR with spatial layout.

9. NASA Algae Raceway Dataset

TIER 4: EXCLUDE FROM PROJECT

Location: `Dataset/1.+Algae+Raceway/algae.zip`
Assessment: Pertains to outdoor biological biomass growth in open raceway ponds (pH, dissolved oxygen, algal density). Unrelated to industrial electromechanical equipment maintenance or troubleshooting.

5. Strategic Implementation Roadmap for Hackathon Demo

1. **Core Telemetry Demo:** Use **AI4I 2020** for real-time fault mode classification in the interactive web UI (uploading telemetry CSVs to instantly flag HDF, TWF, PWF, or OSF).
2. **RUL Prognostics Demo:** Use **NASA C-MAPSS** to showcase Remaining Useful Life degradation curves and prognostic confidence bands.
3. **Multimodal Manual Ingestion:** Use **PubLayNet** + **TableBank** to parse uploaded ABB technical manuals into spatial chunks with bounding boxes.
4. **Field Log & Work Order Ingestion:** Use **FUNSD** + **LayoutLM** (from SROIE) to reliably extract text from noisy, scanned inspection sheets.
5. **Advanced Sensor Deep-Dive:** Utilize **FEMTO Bearing** or **NASA Milling** for vibration spectral analysis and tool degradation wear curves.